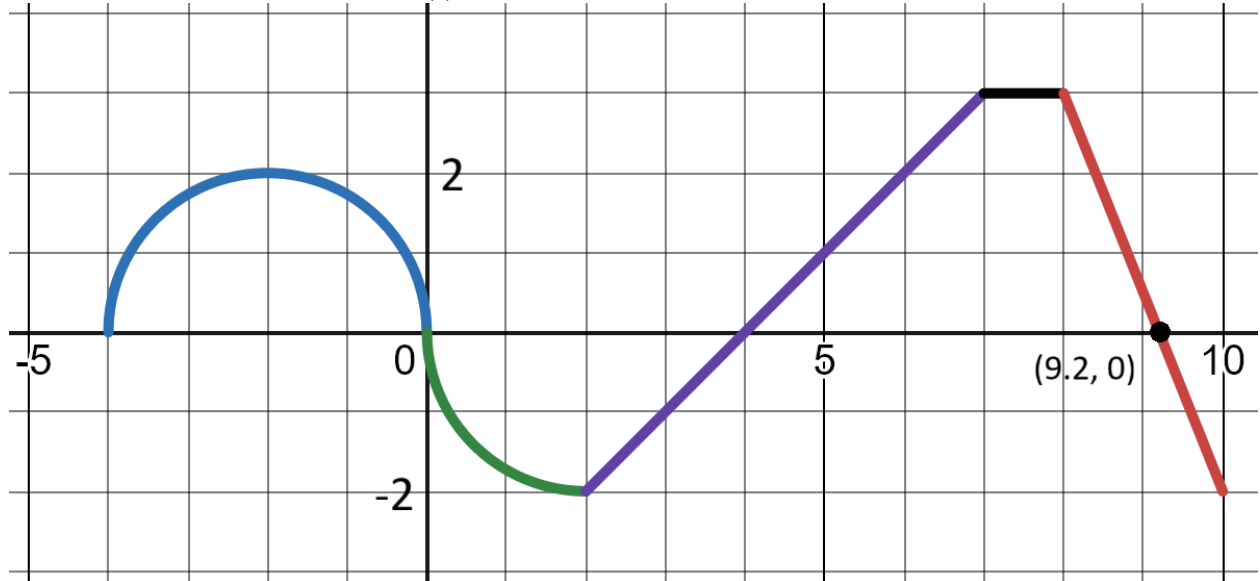


Learning target IN1, version 1

Use geometric formulas to find definite integrals.

Here is the graph of a function $h(t)$, which is made up of only straight lines and parts of circles.



Compute each of the following.

(a) $\int_{-4}^0 h(t) dt$

(e) $\int_7^8 h(t) dt$

(b) $\int_0^2 h(t) dt$

(f) $\int_8^{9.2} h(t) dt$

(c) $\int_2^4 h(t) dt$

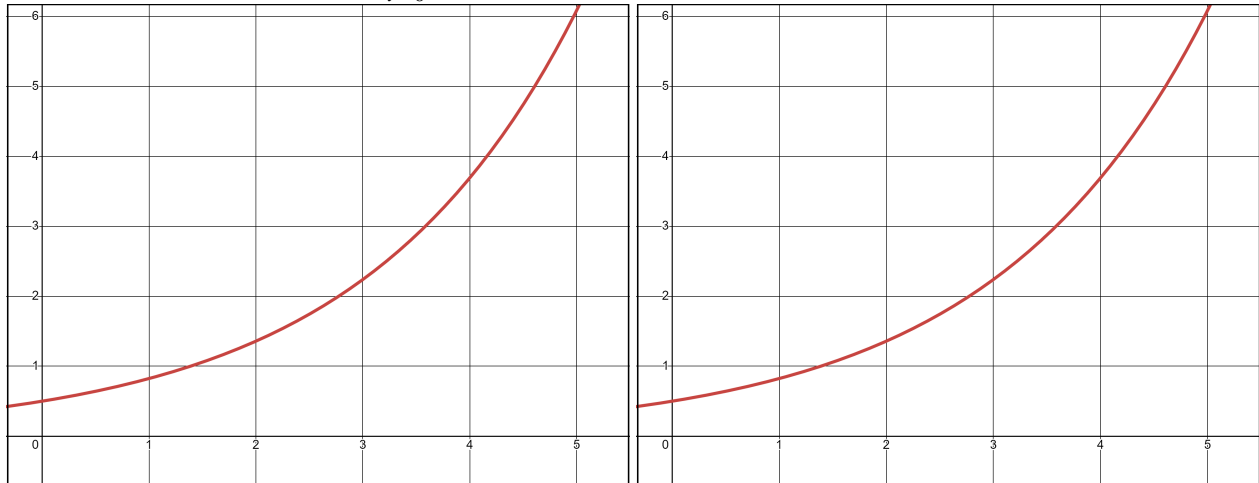
(g) $\int_{9.2}^{10} h(t) dt$

(d) $\int_4^7 h(t) dt$

(h) $\int_{-4}^{10} h(t) dt$

Learning target IN2 and INx, version 1

The rate at which pollution escapes a scrubbing process at a manufacturing plant increases over time as filters and other technologies become less effective. Suppose that the rate of pollution (measured in tons per week) is given by the function r that is pictured below. Throughout this problem, we're interested in $\int_{t=0}^5 r(t) dt$.



- On the first graph above, carefully draw the *left* Riemann sum with four rectangles of uniform width. (How wide should each rectangle be?)
- On the second graph above, carefully draw the *right* Riemann sum with four rectangles.
- If $r(t) = 0.5e^{0.5t}$, compute both of these Riemann sums. Show your work for computing the heights of the rectangles. (Hint: lots of your work will overlap!)

Include units on every number you write.

- Give your best guess for $\int_{t=0}^5 r(t) dt$, **include units**, and explain what this number means about pollution.

Learning target IN3 and IN5, version 1

Evaluate a definite integral using the Fundamental Theorem of Calculus.

Explain how to compute the exact value of each of the following definite integrals using the Fundamental Theorem of Calculus.

Leave all answers in exact form, with no decimal approximations.

(a) $\int_{x=4}^6 \left(\frac{6}{x} \right) dx$

(b) $\int_{x=\frac{5}{4}\pi}^{\frac{4}{3}\pi} (-2 \sec^2(x)) dx$

(c) $\int_{x=-2}^{-1} (3x^3 + 10x - 1) dx$